Research Notes: Sharac Quandles

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1 What is a knot?

A *knot* is a simple closed curve in three-dimensional space, that is, a loop with no loose ends that does not intersect itself. The simplest example is the *unknot*, which is an ordinary circle with no crossings. More interesting examples include the trefoil knot and the figure-eight knot, etc. Examples:



Definition 1. *A knot is a continuous embedding*

$$K : S^1 \to \mathbb{R}^3,$$

where S^1 is the unit circle.

1.1 Equivalence of knots

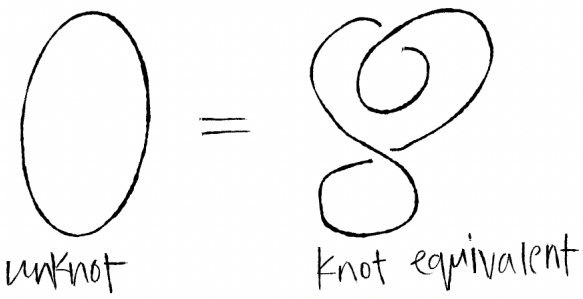
Two knots are considered the same if one can be continuously deformed into the other without cutting the strand or passing it through itself. More precisely, we say two knots K_0 and K_1 are equivalent if there exists a continuous map

$$F : \mathbb{R}^3 \times [0, 1] \to \mathbb{R}^3$$

such that F_0 is the identity, $F_1(K_0) = K_1$, and each F_t is a homeomorphism. This is called *ambient isotopy*.

1.2 Knot diagrams

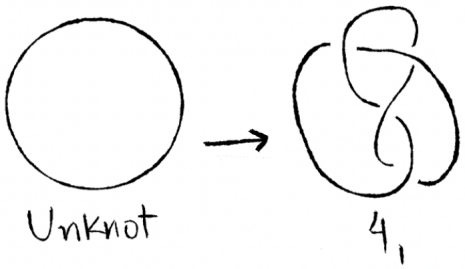
We usually represent knots using *knot diagrams*. These are projections of the knot onto a plane where we indicate which strand passes over and which passes under at each crossing by drawing



a break in the understrand. At each crossing, the strand that goes underneath is broken visually. The pieces of the strand between crossings are called *arcs*. These arcs will become important later when we start assigning algebraic data to a diagram.

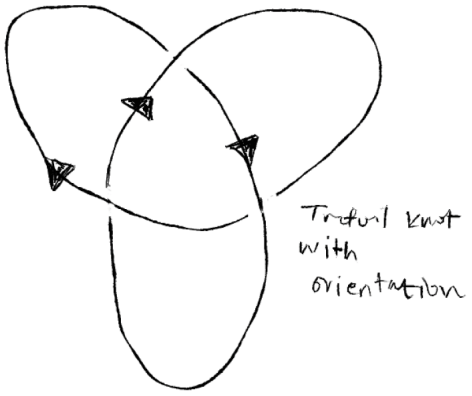
1.3 Reduced

A diagram is called *reduced* if it has the smallest possible number of crossings among all diagrams of the knot.



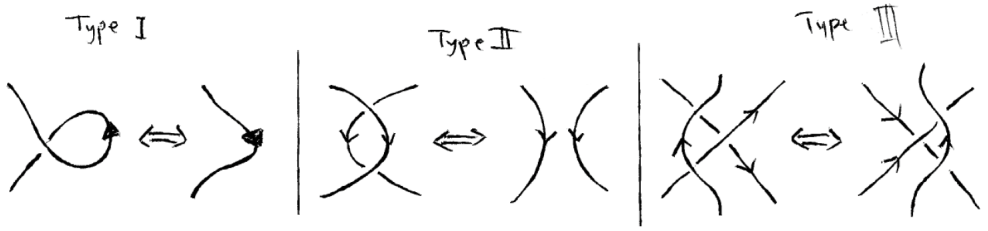
1.4 Orientation of knots

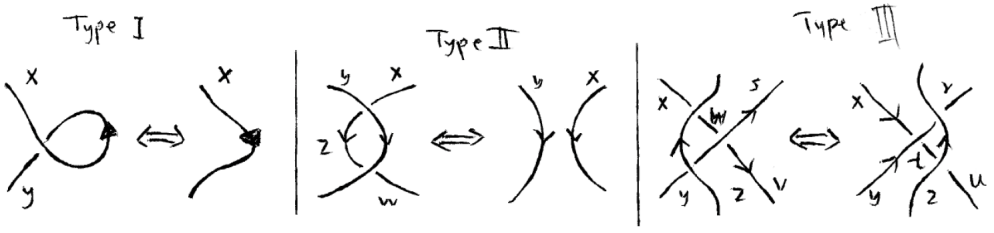
A knot can be given an *orientation*. This means we choose a direction of travel along the curve. We indicate this in a diagram by placing arrows on the strands. Once we choose an orientation, it is carried along consistently around the entire knot. Orientation matters for the coloring of knots by quandles. When we color a knot diagram, we assign elements of a quandle to each arc of the diagram, where an arc is a strand that runs between two consecutive undercrossings. At each crossing, the quandle operation determines how the colors of the incoming strands relate to the colors of the outgoing strands. The orientation of the knot tells us which strand is the overcrossing and which is the undercrossing, and which direction the strands are traveling, so without it the coloring rule is not well defined.



1.5 Reidemeister Moves

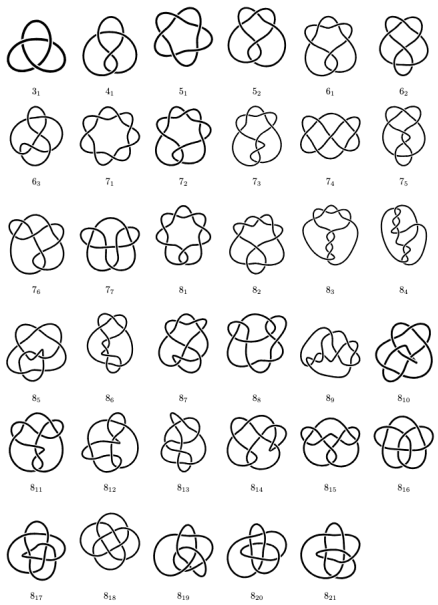
A single knot can be drawn in many different ways. If you take a loop of string and project it onto a flat surface from different angles, you will get different diagrams, but they all represent the same knot. So the question becomes: when do two diagrams represent the same knot? The answer comes from the *Reidemeister moves*, which are three simple local changes you can make to a knot diagram without changing the ambient isotopy of knot itself. Reidemeister proved that two knot diagrams represent the same knot if and only if one diagram can be transformed into the other by a finite sequence of these three moves. This result reduces the question of knot equivalence, which is a geometric question about curves in three-dimensional space, to a combinatorial question about diagrams. This observation is also the key motivation for quandles.





1.6 Knot table

The Tait–Rolfsen knot table lists all prime knots up to ten crossings.



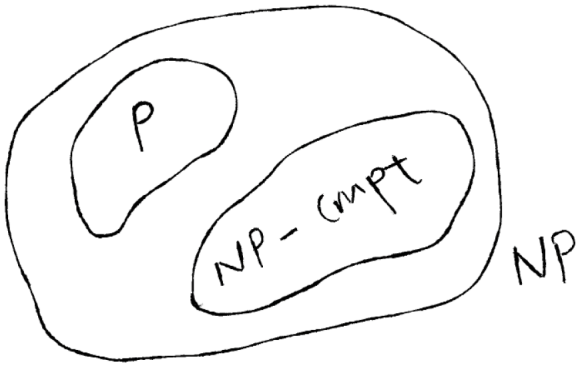
2 What is NP-Completeness

A problem is in the class P if there exists an algorithm that solves it in polynomial time, meaning the time it takes grows like a polynomial function in the size of the input. Problems in P are considered efficiently solvable. For example, sorting a list of numbers is in P.

A problem is in the class NP if, given a proposed solution, we can verify whether it is correct in polynomial time. This is weaker than being able to find the solution efficiently, it only says we can check answers quickly.

Every problem in P is also in NP

A problem is called *NP-complete* if it is in NP and every other problem in NP can be reduced to it in polynomial time. This means NP-complete problems are in some sense the hardest problems in NP. If you could solve any one NP-complete problem efficiently, you could solve all of them efficiently, which would prove P equals NP.



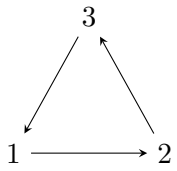
Given a knot K and a finite quandle Q, the problem of determining whether K admits a coloring by Q can be formulated as a constraint satisfaction problem. For certain families of quandles, this problem is NP-complete. $Sharac(n)$ quandles are one such family, which is what gives them their computational significance in knot classification.

3 What is the symmetric group

For a positive integer n, the *symmetric group* S_n is founded over the set of all bijections from the set $\{1, 2, \ldots, n\}$ to itself. These bijections are called *permutations*. The group operation is composition of functions, and the identity element is the permutation that fixes every element.

The group S_n has $n!$ elements. For example, S_3 has $3! = 6$ elements, corresponding to the six ways to rearrange three objects.

Every permutation in S_n can be written in *cycle notation*. A cycle $(a_1\ a_2\ \ldots\ a_k)$ is a permutation that sends a_1 to a_2, a_2 to a_3, and so on, with a_k sent back to a_1. Any element not appearing in the cycle is fixed.



For example, in (1 2 4), the element 1 is sent to 2, 2 is sent to 4, and 4 is sent back to 1

Permutation	**Cycle Notation**	**Cycle Type**
identity	()	$(1, 1, 1)$
swap 1 and 2	(1 2)	$(2, 1)$
swap 1 and 3	(1 3)	$(2, 1)$
swap 2 and 3	(2 3)	$(2, 1)$
rotate 1,2,3	(1 2 3)	(3)
rotate 1,3,2	(1 3 2)	(3)

4 Cycle Types and Conjugacy Classes

Two permutations in S_n are conjugate if and only if they have the same cycle type. Each conjugacy class consists of all permutations with a fixed cycle type, and these classes partition the group. For example, in S_4 the conjugacy classes are:

- the identity, with cycle type $(1, 1, 1, 1)$
- the six transpositions, with cycle type $(2, 1, 1)$
- the eight 3-cycles, with cycle type $(3, 1)$
- the six 4-cycles, with cycle type (4)
- the three products of two disjoint transpositions, with cycle type $(2, 2)$

Cycle Type	**Example**	**Number of Elements**
$(1, 1, 1, 1)$	()	1
$(2, 1, 1)$	(1 2)	6
$(3, 1)$	(1 2 3)	8
(4)	(1 2 3 4)	6
$(2, 2)$	(1 2)(3 4)	3

5 Quandles

Quandles are algebraic structures that were introduced to study knots.

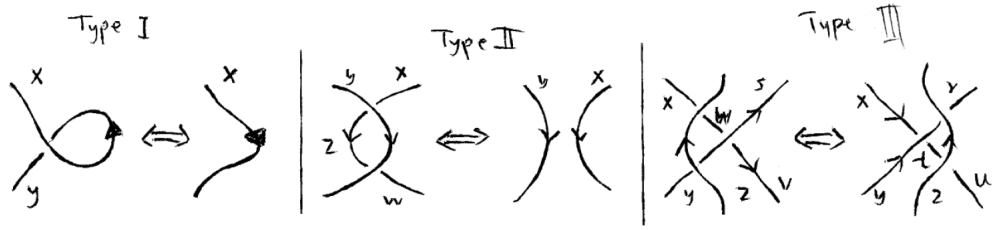
Definition 2. *A quandle is a set Q together with two binary operations $*$ and $/$ satisfying the following axioms for all $a, b, c \in Q$.*

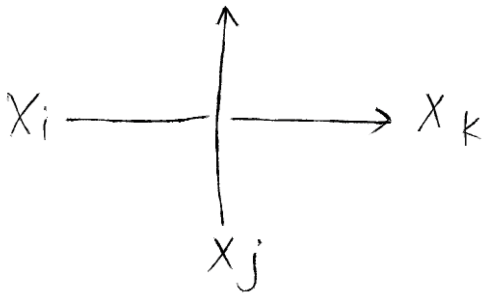
1. ***Idempotence:*** $a * a = a$

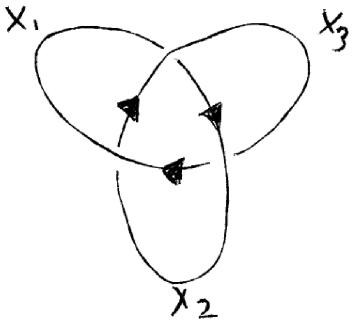
2. ***Right invertibility:*** $(a * b)/b = a$ *and* $(a/b) * b = a$

3. ***Right self-distributivity:*** $(a * b) * c = (a * c) * (b * c)$

Each axiom corresponds to one of the three Reidemeister moves. Idempotence corresponds to the Type I move, right invertibility corresponds to the Type II move, and right self-distributivity corresponds to the Type III move.







The operations $*$ and $/$ are related by the right invertibility axioms. In particular, each operation undoes the other on the right

The simplest example is the trivial quandle. Let Q be any set and define

$$a * b = a \quad \text{for all } a, b \in Q.$$

In this case, every element acts on everything by returning itself. Idempotence holds since $a*a = a$. Right invertibility holds because $(a*b)/b = a$ and $(a/b)*b = a$ are both satisfied when the operation always returns the first input. Self-distributivity also holds since both sides reduce to $a * c = a$.

A more interesting family of examples comes from groups.

Definition 3. *Let G be a group. The* conjugation quandle *of G is the quandle with underlying set G and operations*

$$a * b = b^{-1}ab \quad \textit{and} \quad a/b = bab^{-1}.$$

Let us verify that these operations satisfy the quandle axioms. First,

$$a * a = a^{-1}aa = a,$$

so idempotence holds. For right invertibility, we have

$$(a * b)/b = b(b^{-1}ab)b^{-1} = a$$

and

$$(a/b) * b = b^{-1}(bab^{-1})b = a.$$

Finally, right self-distributivity follows from a direct calculation:

$$(a * b) * c = c^{-1}(b^{-1}ab)c = (c^{-1}bc)^{-1}(c^{-1}ac)(c^{-1}bc) = (a * c) * (b * c).$$

Thus the group G becomes a quandle under conjugation. This example will be especially important when $G = S_n$.

6 Trivial Quandles

Definition 4. *A* trivial quandle *on a set Q is the quandle where*

$$a * b = a \quad \textit{for all } a, b \in Q.$$

Trivial quandles are also called *unary quandles*

Definition 5. *Let Q be a quandle. A subset $S \subseteq Q$ is called a* unary subquandle *if*

$$a * b = a \quad \textit{for all } a, b \in S.$$

Equivalently, the operation restricted to S makes S into a trivial quandle.

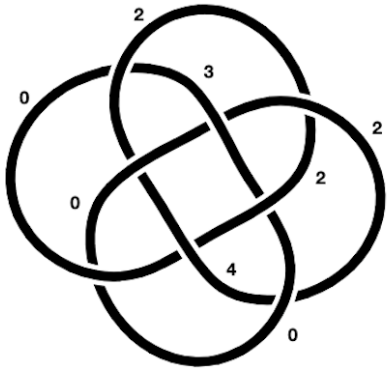
$*$	a	b	c
a	a	a	a
b	b	b	b
c	c	c	c

Table 6.1: Operation table for the trivial quandle on three elements $\{a, b, c\}$

*	0	1	2	3	4	5
0	0	0	3	5	2	4
1	1	1	4	2	5	3
2	4	3	2	0	1	2
3	2	5	1	3	3	0
4	5	2	0	4	4	1
5	3	4	5	1	0	5

/	0	1	2	3	4	5
0	0	0	4	2	5	3
1	1	1	3	5	2	4
2	3	4	2	1	0	2
3	5	2	0	3	3	1
4	2	5	1	4	4	0
5	4	3	5	0	1	5

The operations $*$ and $/$ on $Sharac(4)$



A full coloring of the knot 8_{18} by $Sharac(4)$

$$\mathbf{0 * 2 = 3} \qquad \mathbf{2 * 0 = 4}$$

$$\mathbf{3/2 = 0} \qquad \mathbf{4/0 = 2}$$

$$\mathbf{0 * 0 = 0} \qquad \mathbf{2 * 2 = 2}$$

$$\mathbf{0/3 = 2} \qquad \mathbf{2/4 = 0}$$

Some of the coloring equations satisfied by the $Sharac(4)$ coloring of 8_{18}

7 Algebraic Properties

Definition 6. *A* subquandle *of a quandle Q is a subset $S \subseteq Q$ that is closed under the quandle operation, meaning $a * b \in S$ whenever $a, b \in S$.*

Definition 7. *A quandle Q is called* connected *if the group generated by the right multiplication maps $R_a : x \mapsto x * a$ acts transitively on Q.*

Definition 8. *Let Q be a quandle and let $S \subseteq Q$ be a subquandle. We say that S is a* proper connected subquandle *of Q if $S \neq Q$ and S is connected as a quandle.*

8 Idempotence

The idempotence axiom $a * a = a$ says that every element of a quandle acts trivially on itself.

9 Quandle Homomorphisms

Definition 9. *A* quandle homomorphism *from a quandle Q to a quandle P is a map $f : Q \to P$ such that*

$$f(a * b) = f(a) * f(b)$$

and

$$f(a/b) = f(a)/f(b)$$

for all $a, b \in Q$.

10 Definition of $Sharac(n)$ Quandles

$Sharac(n)$ quandles are defined axiomatically as follows.

Definition 10. *A* $Sharac(n)$ quandle *is a quandle Q satisfying the following conditions:*

1. *Q is a subquandle of a conjugation quandle of a symmetric group.*
2. *Q is connected.*
3. *Q is generated by two elements.*
4. *Q has no nontrivial proper connected subquandles.*
5. *For every $x \in Q$ there exists a nontrivial unary subquandles U_x that contains x. Furthermore there exists an integer $1 < q < n$ with $gcd(q, n) = 1$ and such that*

   $$U_x = \{x, x^q, x^{q^2}, \ldots\}$$

   for all $x \in Q$.
6. *The cycle type of Q only has 1-cycles and n-cycles.*
7. *Let m and k be the number of 1- and n-cycles, respectively, then $n = k + m + 1$*

The first condition places $Sharac(n)$ inside a conjugation quandle, so the operation is inherited from conjugation of permutations. The connectedness and two-generation conditions say that the quandle is not built out of several unrelated pieces and can be generated from a small amount of data. The condition on proper connected subquandles says that $Sharac(n)$ is minimal in the sense that it does not contain a smaller connected subquandle of the same kind.

The last three conditions describe the special cycle structure that makes these quandles useful in computations. In particular, each element has only fixed points and n-cycles in its cycle decomposition, and the relation

$$n = k + m + 1$$

places a strong restriction on which cycle types can occur.

Remark 1. *The relation $k + m + 1 = n$ is taken as part of the defining structure here, rather than as a derived statement. It gives a combinatorial constraint on the allowed cycle decompositions of the elements of Q.*

11 Size Patterns and Structural Results

We can look at the size patterns that appear in the known examples of $Sharac(n)$. The goal is to explain how the cycle structure of the right translation maps gives a formula for the number of elements, and then to describe the different behavior that appears for prime and composite values of n.

12 Derivation of the Size Formula

Let $Q = Sharac(n)$, and let

$$r_x : Q \to Q$$

be the right translation map defined by

$$r_x(y) = y * x.$$

By the defining properties of $Sharac(n)$, the cycle decomposition of each right translation map consists only of 1-cycles and n-cycles.

Suppose that, for a fixed $x \in Q$, the permutation r_x has:

- k cycles of length n,
- m cycles of length 1.

Each cycle of length n contains n elements of Q. Since there are k such cycles, these cycles account for nk elements. Each 1-cycle contributes one fixed element, and there are m of them. Therefore,

$$|Q| = nk + m.$$

The defining relation

$$k + m + 1 = n$$

allows us to rewrite this formula in terms of n and k. Since

$$m = n - k - 1,$$

we have

$$|Q| = nk + (n - k - 1).$$

Thus

$$|Q| = nk + n - k - 1$$

and therefore

$$|Q| = n(k + 1) - k - 1.$$

This formula is useful because different choices of k and m lead to different size patterns. In the examples computed so far, the values of k and m appear to depend strongly on whether n is prime or composite.

13 Prime and Composite Behavior

The equation

$$|Q| = nk + m$$

together with the axiom

$$k + m + 1 = n$$

yields

$$|Q| = n(k + 1) - k - 1.$$

Empirical study of known $Sharac(n)$ quandles suggests distinct behavior depending on whether n is prime or composite.

For many prime values of n, the size appears to simplify to

$$|Sharac(n)| = \frac{n^2 - 1}{2}.$$

For certain composite values of n, computational evidence suggests instead

$$|Sharac(n)| = n(n - 3) + 2 = (n - 1)(n - 2).$$

Understanding why prime and composite values yield different formulas remains an open problem. However, in the case where n is an odd prime, there is a partial result that limits the possible values of k and m.

14 A Divisibility Result for Prime n

Definition 11. *Let Q be a quandle and $x \in Q$ define the* **right translation functions** $rs_x : Q \to Q$ *by $rs_x(y) = y * x$ and $rd_x : Q \to Q$ by $rd_x(y) = y/x$.*

Both rs_x and rd_x are permutations since $rs_x \circ rd_x$ and $rd_x \circ rs_x$ are both the identity function on Q.

The maps rs_x and rd_x are inverse to each other because

$$rs_x(rd_x(y)) = (y/x) * x = y$$

and

$$rd_x(rs_x(y)) = (y * x)/x = y.$$

Therefore, rs_x and rd_x are permutations of the finite set Q.

Theorem 1. *Let k and m be the number of n- and 1-cycles, respectively, in Sharac(n) for n an odd prime. Then $m|k$.*

Some technical results are necessary in order to establish Theorem 1.

Lemma 1. *Let $z \in Sharac(n)$ then for any $y_1, y_2 \in U_z$ there exists a positive integer l with $y_2 = y_1^{q^l}$ or $y_1 = y_2^{q^l}$.*

Proof. by definition of U_z, there exists $i \neq j$ such that $y_1 = x^{q^i}$ and $y_2 = z^{q^j}$. If $i < j$ then $y_2 = y_1^{q^l}$ where $l = j - i$. Otherwise, $y_1 = y_2^{q^l}$ where $l = i - j$. □

Lemma 2. *Let n be an odd prime and $x, z \in Sharac(n)$. Also, let rs_x be the $*$ right translation permutation of Sharac(n). Then for any distinct $y_1, y_2 in U_z$, y_1 and y_2 inhabit distinct cycles of rs_x.*

Proof. Suppose that y_1 and y_2 are both elements of the same cycle π of rs_x. Then since rs_x only has 1-cycles and n-cycles, π is an n cycle. Then for $1 < i < n$, $y_2 = rs_x^i(y_1)$. Also, since $y_1, y_2 \in U_z$, $y_1 * y_2 = y_1$ and $y_2 * y_1 = y_2$. Then $y_2 * rs_z^i(y_2) = rs_z^i(y_1) * rs_z^i(y_2) = rs_z^i(y_1 * y_2) = rs_z^i(y_1) = y_2$ and $rs_z^i(y_2) * y_2 = rs_z^i(y_2) * rs_z^i(y_1) = rs_z^i(y_2 * y_1) = rs_z^i(y_2)$. This means that $rs_z^i(y_2) \in U_{y_2} = U_{y_1}$. Continuing in this fashion yields distinct $y_1, rs_z^i(y_1), rs_z^{2i}(y_1), \ldots, rs_z^{ni} \in U_{y_1}$, since n is prime. This means that $n \leq m$, which is impossible since $n = k + m + 1$ and $k > 0$. □

Lemma 3. *Suppose $x \in Sharac(n)$ and rs_x is the $*$ right translation of x. Suppose that $y_1, y_2 \in Sharac(n)$ are both contained in the same cycle of rs_x. Then if $x^i, y^i \in Sharac(n)$, y_1^i and y_2^i are also in the same cycle of rs_x.*

Proof. Since y_1, y_2 are in the same cycles of rs_x, there exists a nonnegative integer j such that $rs_z^j(y_1) = y_2$. Then

$$rs_x^j(y_1^i) = x^{-j} y_1^i x^i = x^{-j} rs_x(y_1) x^j x^{-j} rs_x(y_1) x^j \ldots x^{-j} rs_x(y_1) x^j = (rs_x^j(y_1))^i = y_2^i.$$

Hence, y_1^i and y_2^i are in the same cycle of rs_x. □

Proof. (Theorem 1) Let $x \in Sharac(n)$ and consider the right translation permutation rs_x. Let C^x be the set of k n-cycles of rs_x. Choose $\pi_1 \in C^x$ and $y_1 \in \pi_1$. By Lemma 2, each of the m elements of $U_{y_1} = \{y_1, y_1^q, y_1^{q^2}, \dots\}$ must inhabit distinct elements of C^x. Let C_1^x be those m cycles of C^x. Next, choose $\pi_2 \in C^x - C_1^x$, and $y_2 \in \pi_2$. Note that by Lemma 3, for every $z \in U_{y_2}$ and $\pi \in C_1^x$, $z \notin \pi$. Form C_2^x from y_2 in the same way that C_1^x was formed from y_1. Then $C_2^x \cap C_1^x = \emptyset$. Continue this construction until C^x is exhausted to find $y_1, y_2, \dots, y_l$ and $C_1^x, C_2^x, \dots, C_l^x$ such that for each $i = 1, 2, \dots, l$, $C_i^x \cap (C_1^x \cup C_2^x \cup \dots \cup C_{i-1}^x) = \emptyset$. Then $\{C_1^x, C_2^x, \dots, C_l^x\}$ forms a partition of C^x of sets of size m. Hence, $m|k$. □

Conjecture 1. *For all sufficiently large prime values of n, the quandle $Sharac(n)$ exists and satisfies*

$$|Sharac(n)| = \frac{n^2 - 1}{2}.$$

The computations also suggest that the composite case behaves differently.

Conjecture 2. *For composite values of n, the quandles $Sharac(n)$ exhibit structurally different size behavior from the prime case.*

In the composite examples studied so far, the size appears to follow the formula

$$|Sharac(n)| = (n-1)(n-2).$$

However, more examples are needed before this pattern can be stated as a general conjecture with confidence.